GPUs AND COLLING SYSTEMS

1.Open your own laptop or desktop (or use an online PC hardware simulator if you don't have access to a real system) and identify the GPU model installed. Write down the GPU's brand, model number, and whether it is integrated or dedicated.

ANS:

- **Grand:** NVIDIA
- **Model:** GeForce GTX 1650
- **Type:** Dedicated (Discrete) Graphics Card

Observation:
The system has an **NVIDIA GeForce GTX 1650** GPU installed. It is a **dedicated (discrete)** graphics card, which is designed to provide better graphics performance for gaming, video editing, and other graphics-intensive applications.

2.Take a clear photo or screenshot of the cooling system (fan, heatsink, or liquid cooling setup) inside your PC or from a PC hardware simulator. Label the main cooling components and briefly describe their function.

ANS:

1. **Cooling Fan**
    - Draws cool air into the case and blows hot air away from the CPU heatsink.
    - Helps maintain safe operating temperatures.
2. **Heatsink**
    - A metal component (usually aluminum or copper) attached to the CPU.
    - Absorbs heat from the processor and transfers it to the surrounding air.
3. **Liquid Cooling Components (if present)**
    - **Water Block/Pump:** Transfers heat from the CPU into the cooling liquid.
    - **Radiator:** Releases the heat carried by the liquid.
    - **Fans:** Blow air through the radiator to cool the liquid before it returns to the CPU.

3.Create a comparison table listing at least 3 differences between air cooling and liquid cooling systems for GPUs, including factors like cost, efficiency, and maintenance.

ANS:

Feature	Aire cooling	Liquid cooling
1.Cost	Lower cost	Higher cost
2.Cooling efficiency	Good for normal use	Better for heavy gaming and overthinking
3.Noise level	Can be louder under heavy load	Usually quieter

4.Imagine you are building a gaming PC for streaming IPL matches and playing graphics-heavy games. Choose between air cooling and liquid cooling for your GPU, and justify your choice in 3-4 sentences based on your research.

ANS:

Choice: Liquid Cooling

I would choose liquid cooling for my gaming PC. It keeps the GPU cooler during long gaming sessions and while streaming IPL matches at the same time. It also runs more quietly than most air coolers under heavy load. Although it is more expensive, its better cooling performance makes it a good choice for high-end gaming.